

**CLASSROOM CONTACT PROGRAMME**
(Academic Session : 2019 - 2020)NEET(UG)
PART TEST-4
01-03-2020**PRE-MEDICAL : SPARK COURSE (PHASE - I)***This Booklet contains 20 pages.***Do not open this Test Booklet until you are asked to do so.****Read carefully the Instructions on the Back Cover of this Test Booklet.****Important Instructions :**

1. On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are 720.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

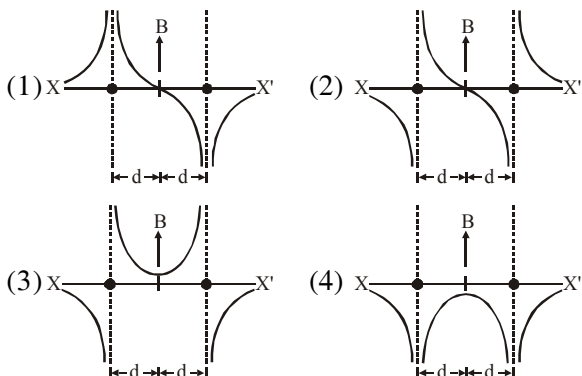
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Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

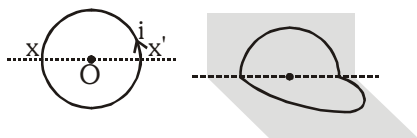
Your Target is to secure Good Rank in Pre-Medical 2020

1. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper, as shown in figure. The variation of the magnetic field B along the line XX' is given by:

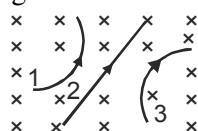


2. In the given figure magnetic field at the centre of ring (O) is $8\sqrt{2}$ T. Now it is through 90° about XX' axis, so that two semicircular parts are mutually perpendicular. Then find the value of magnetic field (in Tesla) at centre :-

- (1) 8
(2) $8\sqrt{2}$
(3) 4
(4) $8/\sqrt{2}$



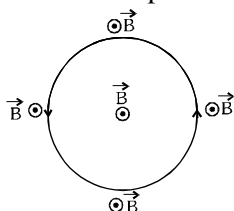
3. The particles 1, 2, 3 are moving in uniform transverse magnetic field then :-



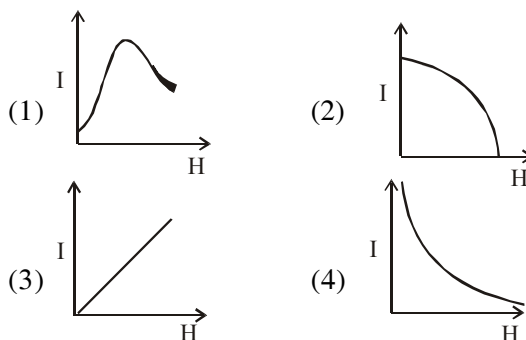
- (1) particle '1' positive and particle 3 negative
(2) particle 1 negative and particle 3 positive
(3) particle 1 negative and particle 2 neutral
(4) particle 1 and 3 are positive and particle 2 neutral

4. An elastic circular wire of length ℓ carries a current I_0 . It is placed in a uniform magnetic field $\vec{B}$ (out of paper) such that its plane is perpendicular to the direction of $\vec{B}$. The wire will experience :-

- (1) No force
(2) A stretching force
(3) A compressive force
(4) A torque



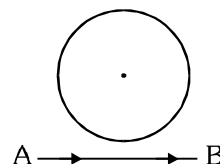
5. The correct I-H curve for paramagnetic materials is:



6. A frog can be levitated in a magnetic field produced by a current in a vertical solenoid placed below the frog. This is possible because the body of the frog behaves as :-

- (1) paramagnetic. (2) diamagnetic
(3) ferromagnetic (4) antiferromagnetic

7. A charge particle moves along the line AB, which lies in the same plane of a circular loop of conducting wire as shown in the fig. Then :-

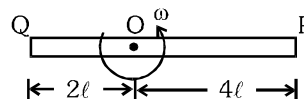


- (1) No current will be induced in the loop
(2) The current induced in the loop will change its direction as the charged particle passes by
(3) The current induced will be anticlockwise
(4) The current induced, will be clockwise

8. A circular loop of radius r is placed in a region where magnetic field increases with respect to time as $B(t) = at$ then induced emf in coil :-

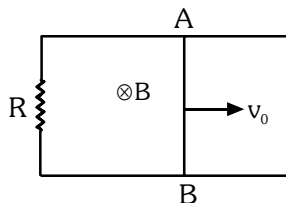
- (1) $\pi r^2 a$ (2) $3\pi r^2 a$ (3) $2\pi r^2 a$ (4) $4\pi r^2 a$

9. A conducting rod rotates with a constant angular velocity ' ω ' about the axis which passes through point 'O' and perpendicular to its length. A uniform magnetic field ' B ' exists parallel to the axis of the rotation. Then potential difference between the two ends of the rod is :



- (1) $6B\omega\ell^2$ (2) $B\omega\ell^2$ (3) $10B\omega\ell^2$ (4) Zero

10. Two long parallel metallic wires with a resistance 'R' form a horizontal plane. A conducting rod AB is on the wires shown in figure. The space has magnetic field pointing vertically downwards. The rod is given an initial velocity ' v_0 '. There is no friction in the wires and the rod. After a time 't' the velocity v of the rod will be such that:-



- (1) $v > v_0$ (2) $v < v_0$ (3) $v = v_0$ (4) $v = -v_0$

11. The armature coil of dynamo is rotating. The generated induced emf varies and the number of magnetic lines of force also varies. Which of the following condition is correct:-

- (1) lines of flux will be minimum, but induced emf will be zero.
 (2) lines of flux will be maximum, but the induced emf will be zero.
 (3) lines of flux will be maximum, but induced emf will be not be zero.
 (4) the lines of flux will be maximum, and the induced emf will be also maximum.

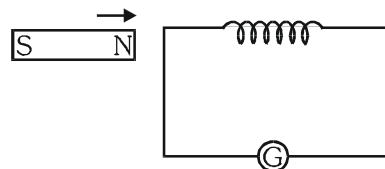
12. In an AC generator, a coil with N turns, all of the same area A and total resistance R , rotates with frequency ω in a magnetic field B . The maximum value of emf generated in the coil is :-

- (1) $NAB\omega$ (2) $NABR\omega$ (3) NAB (4) $NABR$

13. A coil of resistance 400Ω is placed in a magnetic field. If the magnetic flux ϕ (Wb) linked with the coil varies with time t (sec) as $\phi = 50t^2 + 4$. The current in the coil at $t = 2$ sec is :

- (1) 2A (2) 1A (3) 0.5A (4) 0.1A

14. As shown is the figure, a magnet is brought towards a fixed coil. Due to this the induced emf, current and the charge are E , I and Q respectively. If the speed of the magnet is double then the following statement is wrong :-

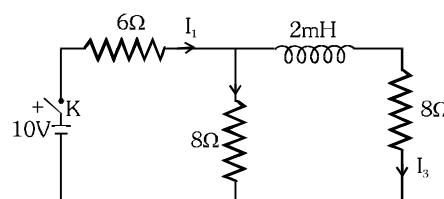


- (1) E increases (2) I increases
 (3) Q does not change (4) Q increases

15. A solenoid having 500 turns and length 2 m has radius of 2 cm, then self inductance of solenoid?

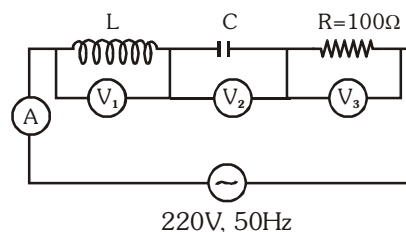
- (1) 4×10^{-4} H (2) 2×10^{-4} H
 (3) 8×10^{-4} H (4) 16×10^{-4} H

16. In the circuit shown in figure what is the value of I_1 just after pressing the key K ?



- (1) $\frac{5}{7}$ A (2) $\frac{5}{11}$ A
 (3) 1A (4) None of the above

17. In the given circuit the reading of voltmeter V_1 and V_2 are 300 volts each. The reading of the voltmeter V_3 and ammeter A are respectively :



- (1) 100 V, 2.0 A (2) 150 V, 2.2 A
 (3) 220 V, 2.2 A (4) 220 V, 2.0 A

18. In an ac circuit an alternating voltage $e = 200\sqrt{2} \sin 100t$ volts is connected to a capacitor of capacity $1\mu\text{F}$. The r.m.s. value of the current in the circuit is :-

- (1) 10 mA (2) 100 mA
 (3) 200 mA (4) 20 mA

19. If frequency of alternating source is made zero then which of the following statement is true :

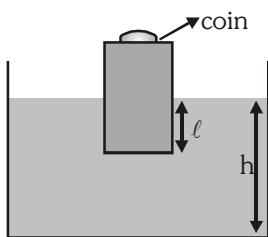
- (1) Current through capacitor will be zero
- (2) Current through resistance will be zero
- (3) Current through inductance will be zero
- (4) All

20. An object of weight W and density ρ is submerged in a fluid of density ρ_1 . Its apparent weight will be (Given : $\rho_1 < \rho$)

(1) $W(\rho - \rho_1)$ (2) $\frac{(\rho - \rho_1)}{W}$

(3) $W\left(1 - \frac{\rho_1}{\rho}\right)$ (4) $W(\rho_1 - \rho)$

21. A wooden block, with a coin placed on its top, floats in water as shown in figure. The distance ℓ and h are shown there. After sometime the coin falls into the water. Then :-



- (1) ℓ decreases and h increases
- (2) ℓ increases and h decreases
- (3) both ℓ and h increase
- (4) both ℓ and h decrease

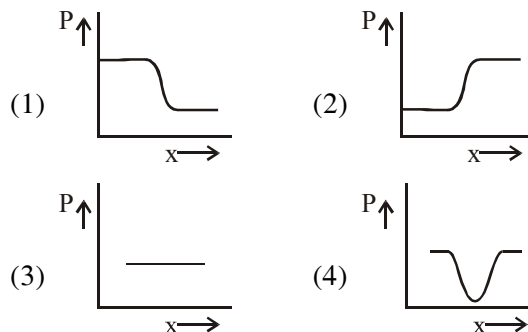
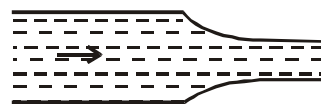
22. Water is flowing through a non-uniform radius tube. If ratio of the radius of entry and exit end of the pipe is 3 : 2 then the ratio of velocities of entering and exit liquid is :-

- (1) 4 : 9 (2) 9 : 4 (3) 8 : 27 (4) 1 : 1

23. Speed of 2 cm radius ball in a viscous liquid is 20 cm/s. Then the speed of 1 cm radius ball in the same liquid is

- (1) 5 cm/s (2) 10 cm/s (3) 40 cm/s (4) 80 cm/s

24. Water flows through a frictionless duct with a cross-section varying as shown in figure. Pressure P at points along the axis is represented by



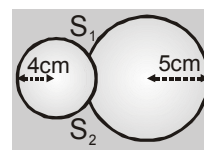
25. A liquid drop of diameter D breaks into 27 tiny drops. The resultant change in energy is :-

- (1) $4\pi TD^2$ (2) $2\pi TD^2$
(3) πTD^2 (4) None of these

26. In a capillary tube, water rises by 1.2 mm. The height of water that will rise in another capillary tube having half the radius of the first, is :

- (1) 1.2 mm (2) 2.4 mm (3) 0.6 mm (4) 0.4 mm

27. Two soap bubbles of radii r_1 and r_2 equal to 4 cm and 5 cm are touching each other over a common surface S_1S_2 (shown in figure). Its radius will be:-



- (1) 4 cm (2) 20 cm (3) 5 cm (4) 4.5 cm

28. The excess pressure inside a soap bubble A is twice that in another soap bubble B. The ratio of volumes of A and B is

- (1) 1 : 2 (2) 1 : 4 (3) 1 : 8 (4) 1 : 16

29. Three liquids of densities ρ_1 , ρ_2 and ρ_3 (with $\rho_1 > \rho_2 > \rho_3$), having the same value of surface tension T , rise to the same height in three identical capillaries. The angles of contact θ_1 , θ_2 and θ_3 obey:-

- (1) $\frac{\pi}{2} < \theta_1 < \theta_2 < \theta_3 < \pi$ (2) $\pi > \theta_1 > \theta_2 > \theta_3 > \frac{\pi}{2}$
(3) $\frac{\pi}{2} > \theta_1 > \theta_2 > \theta_3 \geq 0$ (4) $0 \leq \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$

30. An air bubble is lying just below the surface of water. The surface tension of water is $70 \times 10^{-3} \text{ N/m}$ and atmospheric pressure is $1.013 \times 10^5 \text{ N/m}^2$. If the radius of bubble is 1 mm, then the pressure inside the bubble will be :-

- (1) $1.0270 \times 10^5 \text{ Pa}$ (2) $1.0160 \times 10^5 \text{ Pa}$
(3) $1.0144 \times 10^5 \text{ Pa}$ (4) $1.0131 \times 10^5 \text{ Pa}$

31. Water stands upto a height h behind the vertical wall of a dam. What is the net horizontal force pushing the dam down by the stream, if width of the dam is σ ? (ρ = density of water)

- (1) $2h\sigma g$ (2) $\frac{h^2\sigma\rho g}{2}$ (3) $\frac{h^2\sigma\rho g}{4}$ (4) $\frac{h\sigma\rho g}{4}$

32. To what height h should a cylindrical vessel of diameter d be filled with a liquid so that due to liquid force on the vertical surface of the vessel be equal to the force on the bottom :-

- (1) $h = d$ (2) $h = 2d$ (3) $h = 3d$ (4) $h = d/2$

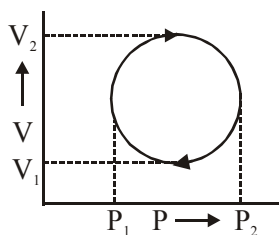
33. The gravitational potential energy of a body at a distance r from the center of the earth is U . The force at that point is :

- (1) $\frac{|U|}{r^2}$ (2) $\frac{|U|}{r}$ (3) $|U|r$ (4) $|U|r^2$

34. The radii of circular orbits of two satellites A and B of the earth, are $4R$ and R , respectively. If the speed of satellite A is $3V$, then the speed of satellite B will be :-

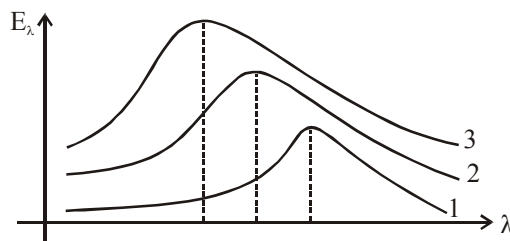
- (1) $3V/2$ (2) $3V/4$ (3) $6V$ (4) $12V$

35. In the cyclic process shown by indicator diagram, the work done on the gas is :-



- (1) $\frac{\pi}{4}(P_2 - P_1)^2$ (2) $\frac{\pi}{4}(V_2 - V_1)^2$
(3) $\pi(P_2V_2 - P_1V_1)$ (4) $\frac{\pi}{4}(P_2 - P_1)(V_2 - V_1)$

36. The graphs shown in the figure represent energy density E_λ versus λ for three sources sun, welding arc and tungsten filament. For λ_{max} , correct combination will be :-

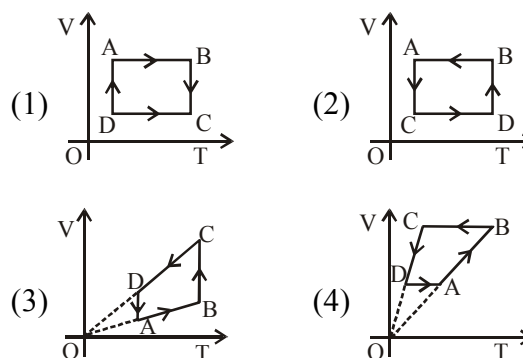
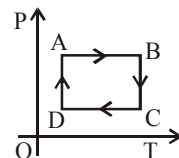


- (1) 1-tungsten, 2-welding arc, 3-sun
(2) 1-sun, 2-tungsten, 3-welding arc
(3) 1-sun, 2-welding arc, 3-tungsten
(4) 1-welding arc, 2-sun, 3-tungsten

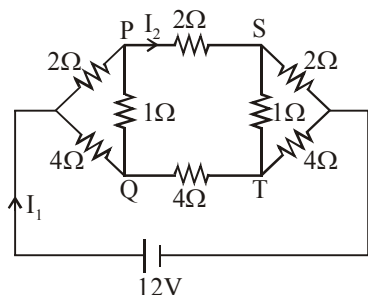
37. A container encloses 32 gm of O_2 at 127°C and of pressure P_0 . A small hole is made in the container and the gas leaks out. At a certain state, pressure of gas becomes $\frac{P_0}{2}$ and its temperature 27°C . Mass of the gas that has leaked out till this stage is nearly:-

- (1) 6.3 gm (2) 7.4 gm
(3) 16.3 gm (4) 10.7 gm

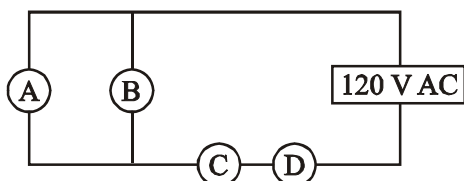
38. A cyclic process is shown on the P-T diagram. Which of the curve shows the same process on a V-T diagram.



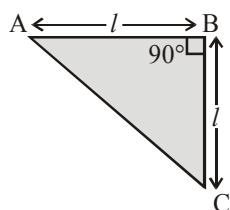
39. For the resistance network shown in the figure, choose the correct option(s):



- (1) The current through 'PQ' is zero
 (2) $I_2 = 2A$
 (3) $I_1 = 3A$
 (4) All of the above
40. A galvanometer has resistance 100Ω and it requires current $100\mu A$ for full scale deflection. A resistor 0.1Ω is connected in parallel to make it an ammeter. The smallest current required in the circuit to produce the full scale deflection is:
- (1) 1000.1 mA (2) 1.1 mA
 (3) 10.1 mA (4) 100.1 mA
41. The diagram shows a circuit with four identical light bulbs.

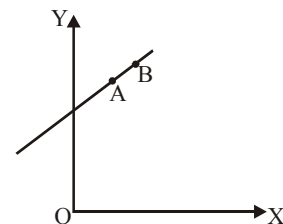


- When we removed bulb A from the circuit, the light intensity of bulb C :
- (1) Remains the same (2) Increases
 (3) Decreases (4) Becomes zero
42. Figure shows a thin metallic triangular sheet ABC. The mass of the sheet is M. The moment of inertia of the sheet about side AC is



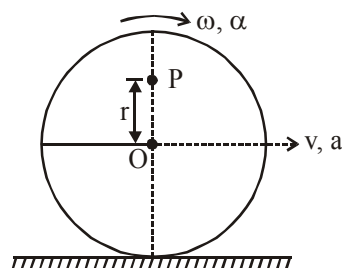
- (1) $\frac{Ml^2}{18}$ (2) $\frac{Ml^2}{12}$ (3) $\frac{Ml^2}{6}$ (4) $\frac{Ml^2}{4}$

43. A particle of mass m moves in the XY plane with a velocity v along the straight line AB. If the angular momentum of the particle with respect to origin O is L_A when it is at A and L_B when it is at B, then :-



- (1) $L_A > L_B$
 (2) $L_A = L_B$
 (3) the relationship between L_A and L_B depends upon the slope of the line AB
 (4) $L_A < L_B$
44. A small object of uniform density rolls up a curved surface with an initial velocity v. It reaches up to a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is
- (1) Ring (2) Solid sphere
 (3) Hollow sphere (4) Disc

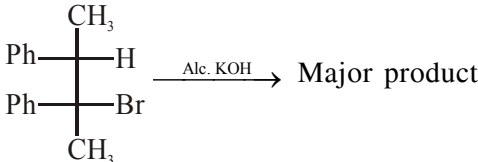
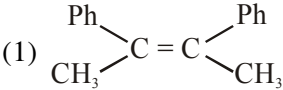
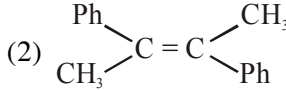
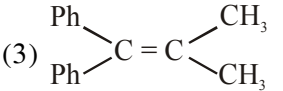
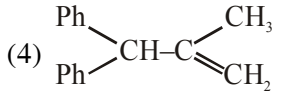
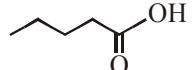
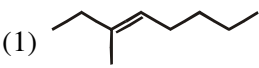
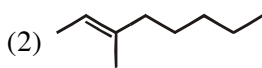
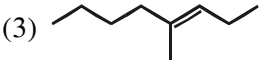
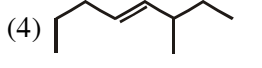
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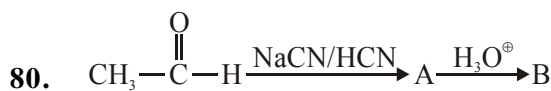


A disc of radius r rolls on a horizontal ground with linear acceleration a and angular acceleration α . The magnitude of acceleration of point P at an instant when its linear velocity is v and angular velocity is ω will be :

- (1) $\sqrt{(a + r\alpha)^2 + (r\omega^2)^2}$ (2) $\frac{ar}{R}$
 (3) $\sqrt{r^2\alpha^2 + r^2\omega^4}$ (4) $r\alpha$

46. Which of the following pairs of elements have almost similar atomic radii :-
 (1) Zr, Hf (2) Mo, W
 (3) Co, Ni (4) All
47. The correct values of ionisation energies (in kJ mol⁻¹) of Be, Ne, He and N respectively are
 (1) 786, 1012, 999, 1256
 (2) 1012, 786, 999, 1256
 (3) 786, 1012, 1256, 999
 (4) 786, 999, 1012, 1256
48. The order of ionisation potential between He⁺ ion and H-atom (both species are in gaseous state) is:-
 (1) I.P. (He⁺) = I.P. (H)
 (2) I.P. (He⁺) < I.P. (H)
 (3) I.P. (He⁺) > I.P. (H)
 (4) Cannot be compared
49. Which is the correct order of 2nd IP.
 (1) Si < S < P < Cl (2) Si < P < S < Cl
 (3) P < Si < Cl < S (4) Si < P < Cl < S
50. Which is correct order for boiling point ?
 (1) H₂O > HCl (2) O₂ < S₈
 (3) F₂ < Cl₂ < Br₂ < I₂ (4) All
51. Which of the radius order is not correct ?
 (1) Ti < Zr = Hf
 (2) B < Al < Ga < In = Tl
 (3) O < N < S < P
 (4) Li < Na < K
52. Correct order of Boiling point is :-
 (1) HF > HI > HBr > HCl
 (2) HI > HF > HCl > HF
 (3) HF > HCl > HBr > HI
 (4) HCl > HBr > HI > HF
53. Which one of the following order of given properties is correct ?
 (1) Atomic radius - Li < Be < B
 (2) Ionisation potential - Li < Be < B
 (3) Electron affinity - Li < Be < B
 (4) Electronegativity - Li < Be < B
54. Which of the following compound is amphoteric in nature ?
 (1) ZnO (2) Al₂O₃ (3) BeO (4) AlI
55. Which has the lowest anion to cation size ratio ?
 (1) LiF (2) NaF (3) CsI (4) CsF
56. Which has fractional bond order :-
 (1) B₂ (2) O₂⁺ (3) C₂ (4) C₂⁺
57. Which of the following compound can form H-bond with water ?
 (1) NH₃ (2) HF (3) NaOH (4) All
58. Which of the following ionisation energy order is correct?
 (1) Cl < F⁻ < Cl⁻ < F (2) F < Cl < Cl⁻ < F⁻
 (3) Cl⁻ < F⁻ < F < Cl (4) F⁻ < Cl⁻ < Cl < F
59. In which of the following move Z_{eff} increases by .65 (according to Slater's rule)?
 (1) B to C (2) Ca to Mg
 (3) Na to K (4) Li to Na
60. Which of the following is an exception of octet rule ?
 (1) BeCl₂ (2) AlF₃ (3) CO₂ (4) NH₃
61. The correct order of bond dissociation energy of halogens is :-
 (1) F₂ > Cl₂ > Br₂ > I₂
 (2) Cl₂ > F₂ > Br₂ > I₂
 (3) Cl₂ > Br₂ > F₂ > I₂
 (4) I₂ > Cl₂ > Br₂ > F₂
62. Which of the following is an odd e⁻ species ?
 (1) Cl₂O₆ (2) N₂O₄ (3) N₂O₃ (4) ClO₃
63. In which of the following compound all the bond angles are equal ?
 (1) SF₄ (2) CCl₄ (3) CHCl₃ (4) XeF₆
64. Increasing order of metallic character will be
 (1) P < Si < Be < Mg < Na
 (2) P > Si > Be > Mg > Na
 (3) P < Si < Be > Mg < Na
 (4) P > Si < Be < Mg < Na
65. Among the following largest and smallest size will be Mg, Mg⁺², Al, Al³⁺
 (1) Mg⁺², Al³⁺ (2) Al³⁺, Mg⁺²
 (3) Al³⁺, Mg (4) Mg, Al³⁺

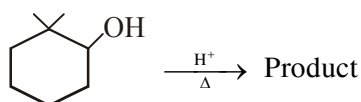
66. Among P, S, Cl, F most and least negative ΔH_{eg} will be of:
 (1) Cl, P (2) P, Cl
 (3) Cl, F (4) F, Cl
67. The strength of bonds by 2s - 2s, 2p - 2p and 2p - 2s overlapping has the order :-
 (1) s - s > p - p > s - p
 (2) s - s > p - s > p - p
 (3) p - p > s - p > s - s
 (4) p - p > s - s > p - s
68. In which of the excitation state of chlorine ClF_3 is formed :-
 (1) In ground state
 (2) In third excitation state
 (3) In first excitation state
 (4) In second excitation state
69. Covalent bond is formed by
 (1) Combination of ions
 (2) Combination of half filled orbitals
 (3) Exchange of electrons by atom
 (4) None of above
70. The correct order of bond length is
 (1) $C - C < C = C < C \equiv C$
 (2) $C \equiv C < C = C < C - C$
 (3) $C = C < C \equiv C < C - C$
 (4) $C = C < C - C < C \equiv C$
71. The d- orbitals involved in sp^3d hybridisation is:-
 (1) $d_{x^2-y^2}$ (2) d_{z^2}
 (3) d_{xy} (4) d_{xz}
72. The shape of IF_4^+ will be :-
 (1) Square planar
 (2) Tetranedral
 (3) Pentagonal bipyramidal
 (4) See-Saw
73. Which of the following has pyramidal shape ?
 (1) BF_3 (2) H_3O^+
 (3) NO_3^- (4) CO_3^{2-}
74. The geometry of XeO_3 is :-
 (1) Triangular planar
 (2) Trigonal bipyramidal
 (3) Pyramidal
 (4) Octahedral
75. Which of the following is having a square planar structure :-
 (1) NH_4^+ (2) BF_4^-
 (3) XeF_4 (4) CCl_4
76. Which of the following molecule have zero dipole moment:-
 (1) BF_3 (2) CH_2Cl_2
 (3) NF_3 (4) SO_2
77. The correct order of dipole moment is :
 (1) $CH_4 < NF_3 < NH_3 < H_2O$
 (2) $NF_3 < CH_4 < NH_3 < H_2O$
 (3) $NH_3 < NF_3 < CH_4 < H_2O$
 (4) $H_2O < NH_3 < NF_3 < CH_4$
78. 
 (1)  (2) 
 (3)  (4) 
79. Alkene (X) $\xrightarrow[H^+]{KMnO_4}$  + 
 X is :-
 (1)  (2) 
 (3)  (4) 

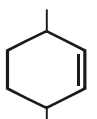
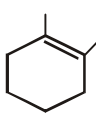
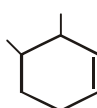
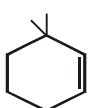


What is the structure of product B ?

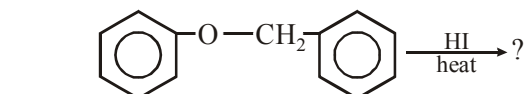
- (1) $\text{CH}_3-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\text{CN}$
 (2) $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\text{COOH}$
 (3) $\text{CH}_3-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{COOH}$
 (4) $\text{CH}_3-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\text{COOH}$

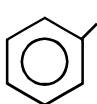
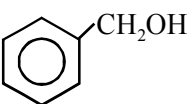
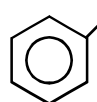
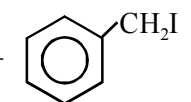
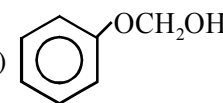
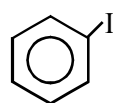
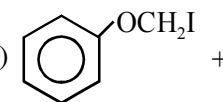
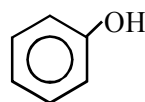
81. Major product in the following reaction is-



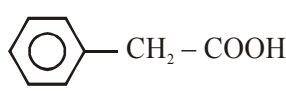
- (1)  (2) 
 (3)  (4) 

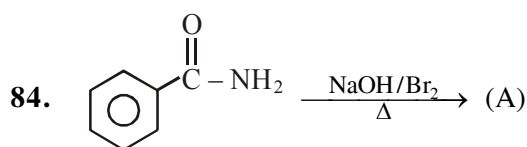
82. What are the major products from the following reaction?



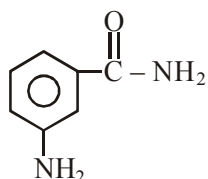
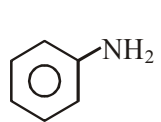
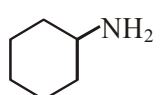
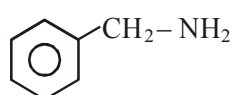
- (I)  + 
 (II)  + 
 (III)  + 
 (IV)  + 
 (1) I (2) II (3) III (4) IV

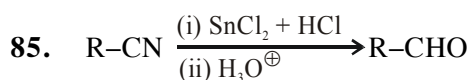
83. Which of the following undergoes decarboxylation with soda lime at maximum rate?

- (1) $\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$
 (2) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{COOH}$
 (3) 
 (4) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{COOH}$



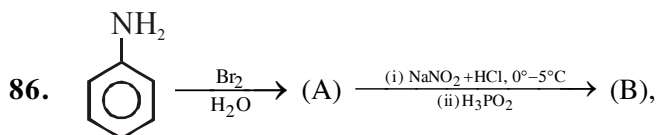
Product A is :-

- (1)  (2) 
 (3)  (4) 

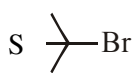
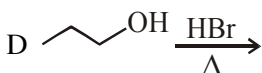
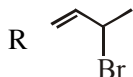
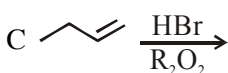
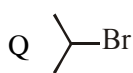
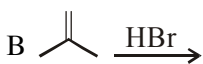
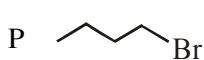
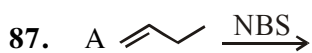
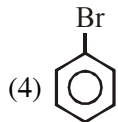
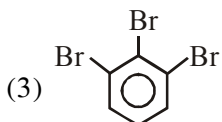
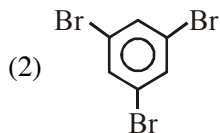
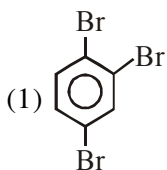


Above reaction is known as :-

- (1) Darzen Reduction
 (2) Rosenmund Reduction
 (3) Stephen Reduction
 (4) Nef Reaction



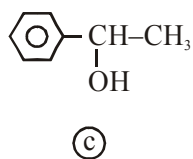
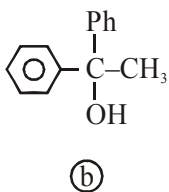
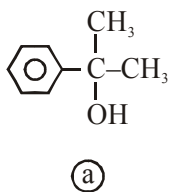
Product (B) in this reaction is :-



Match correct product of given reactions :-

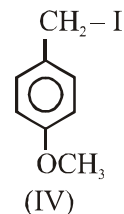
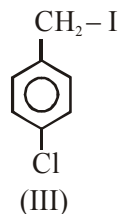
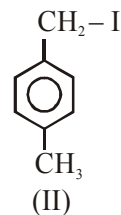
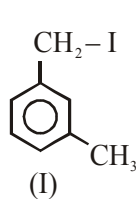
- (1) A $\rightarrow$ P B $\rightarrow$ Q C $\rightarrow$ R D $\rightarrow$ S
 (2) A $\rightarrow$ R B $\rightarrow$ S C $\rightarrow$ P D $\rightarrow$ Q
 (3) A $\rightarrow$ S B $\rightarrow$ R C $\rightarrow$ P D $\rightarrow$ Q
 (4) A $\rightarrow$ R B $\rightarrow$ S C $\rightarrow$ Q D $\rightarrow$ P

88. Compare the rate of dehydration in the given alcohol :-



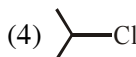
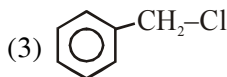
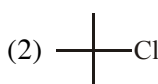
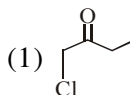
- (1) a > b > c (2) b > c > a
 (3) b > a > c (4) c > b > a

89. Correct order of rate of $\text{S}_\text{N}1$ reaction is :-



- (1) III > I > II > IV
 (2) IV > III > II > III
 (3) IV > II > I > III
 (4) IV > I > II > III

90. Which is more reactive in $\text{S}_\text{N}2$ mechanism :



- | | |
|--|--|
| <p>91. The term systematics refer to :-
 (1) Identification and study of organisms
 (2) Identification and preservation of plants and animals
 (3) Study of habitat of organism and their classification
 (4) Diversity of kind of organisms and their relationship</p> <p>92. In which of the following is not the aspect of taxonomy ?
 (1) Characterisation (2) Identification
 (3) Responsiveness (4) Nomenclature</p> <p>93. Which is not correctly matched with respect to its taxonomic rank ?
 (1) Order - Solanaceae (2) Homo - Genus
 (3) Felidae - Family (4) Mammalia - Class</p> <p>94. Select the incorrect statement :-
 (1) Muscidae is the order of housefly
 (2) Mammalia is the class of human
 (3) Sapindales is the order of mango
 (4) Monocotyledonae is the class of wheat</p> <p>95. The biological concept of species given by Ernst Mayr is based on :-
 (1) Morphological characters
 (2) Floral characters
 (3) Sexual characters
 (4) Sexual reproduction</p> <p>96. Read the statement (A-D) and select the option having all the correct statement :-
 (A) Intrinsic growth is the defining property of living organisms.
 (B) Wheat, plant, Mammal, dog, human etc are taxa.
 (C) The process of classification is known as nomenclature.
 (D) Lower the taxa, lesser are the characteristics that the members. Share with in the taxon.
 (1) Three (2) Four (3) Two (4) One</p> | <p>97. In which of the following statement is incorrect about numerical taxonomy ?
 (1) It is based on observable characters.
 (2) Use of computer.
 (3) More significant in using fossil records.
 (4) Equal weightage is given to each character.</p> <p>98. Natural classification was proposed by :-
 (1) Bentham and Hooker
 (2) Oswald Tippo
 (3) Carolus Linnaeus
 (4) Aristotle</p> <p>99. If a moneran is formed in salty area and synthesis ATP due to presence of a pigmented membrane in the presence of sunlight, the moneran should be a :-
 (1) Rumenococcus (2) Halophiles
 (3) Thermoplasma (4) Nostoc</p> <p>100. Most of pathogenic bacteria are :-
 (1) Bacilli form and spore forming
 (2) Cocci form and spore forming
 (3) Bacilli form and nonspore forming
 (4) Cocci form and nonspore forming</p> <p>101. Select the correct statement :-
 (1) Mycoplasma are insensitive to penicillin.
 (2) Archaeobacteria have cellulosic cell wall
 (3) <i>Frankia</i> perform nitrogen fixation in free living condition
 (4) <i>Rhodospirillum</i> is oxygenic free living nitrogen fixing bacteria.</p> <p>102. Which set of organisms share same kingdom according to five kingdom system of classification?
 (1) <i>Amoeba</i>, <i>Paramecium</i>, <i>Plasmodium</i>
 (2) <i>Gonyaulax</i>, <i>Nostoc</i>, <i>Amoeba</i>, <i>Desmids</i>
 (3) Sulphur bacteria, iron bacteria, <i>Colletotrichum</i>
 (4) <i>Anabena</i>, <i>Euglena</i>, <i>Physarum</i></p> |
|--|--|

103. Read the following statements :-

- (A) Diatoms and Desmids share common kingdom.
- (B) Euglenoids are mostly marine.
- (C) Mycoplasma are obligate aerobes.
- (D) Diatoms are chief producers of ocean.
- (E) Heterocysts in cyanobacteria are helpful in nitrogen fixation

How many of the above statements are correct ?

- (1) All (2) One
- (3) Two (4) Three

104. Which of the following are members of deuteromycetes :-

- (1) *Alternaria*, *Trichoderma*
- (2) *Ustilago*, *Puccinia*
- (3) *Neurospora*, *Claviceps*
- (4) *Penicillium*, Yeast

105. Identify the incorrect information about given fungi :-

- (1) *Puccinia* - it causes wheat rust
- (2) *Neurospora* - extensively used in genetics
- (3) *Claviceps* - source of LSD
- (4) *Ustilago* - it is also known as rust fungi

106. A belong to a class of fungi in which sexual reproduction is absent members of the class produce B, C :-

- | A | B | C |
|---------------------------|-----------|--------------|
| (1) <i>Alternaria</i> | Conidia | Exogenously |
| (2) <i>Alternaria</i> | Zoospores | Endogenously |
| (3) <i>Colletotrichum</i> | Conidia | Endogenously |
| (4) <i>Colletotrichum</i> | Zoospores | Exogenously |

107. Edible fungi like morels and truffles belong to ___ class and are expected to sexually reproduce by producing ___ :-

- (1) Basidiomycetes, conidia
- (2) Basidiomycetes, basidiospores
- (3) Ascomycetes, ascospores
- (4) Ascomycetes, conidia

108. Diplontic life cycle is found in :-

- (1) *Polysiphonia* (2) *Fucus*
- (3) Kelp (4) *Sphagnum*

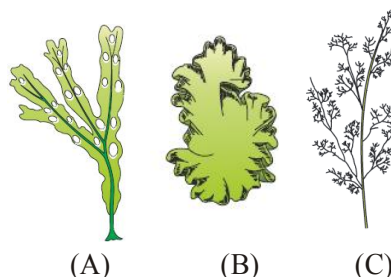
109. Select the incorrect set of statements :-

- (A) Methanogens produce biogas from dung in ruminants.
 - (B) Protistans may have siliceous cell wall.
 - (C) Euglenoids usually possess two equal flagella.
 - (D) Dinoflagellates are usually marine.
 - (E) Mycoplasma are wall less prokaryotes.
- (1) A and C (2) B and E
 - (3) B and C (4) C only

110. Isogamous reproduction is observed in :-

- (1) *Volvox*, *Spirogyra*
- (2) *Spirogyra*, *Fucus*
- (3) *Volvox*, *Fucus*
- (4) *Spirogyra*, *Chlamydomonas*

111. Select the incorrect information given about the algae shown in the given diagrams :-

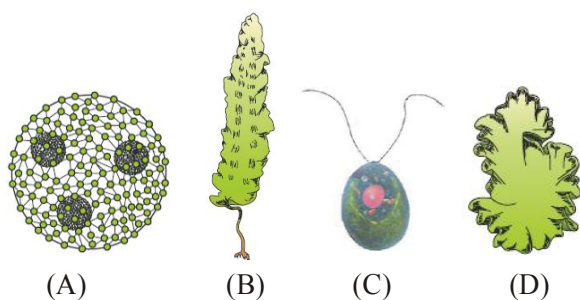


- (1) All of them have cellulose in their cell wall
- (2) Two of them are having floridean starch as stored material
- (3) At least one of them is having chlorophyll-b.
- (4) Two of them are having phycoerythrin as photosynthetic pigment

112. Which of the following set of algae belongs to class phaeophyceae ?

- (1) *Fucus*, *Dictyota*
- (2) *Polysiphonia*, *Gelidium*
- (3) *Chara*, *Spirogyra*
- (4) *Ulothrix*, *Fucus*

113. Diagram of four algae are given below. Select the correct statement following the diagram :-



- (1) A and B belong to same group and have cellulose in their cell walls.
- (2) B and D belongs to different groups and both have cellulose and algin in their cell walls.
- (3) A and C belongs to same group and both have chlorophyll a and b as major photosynthetic pigments.
- (4) C and D belongs to different groups and both have floridean starch as stored food material.

114. A and B are two bryophytes. Gametophyte of A is thalloid with gemmae while that B is having two stages protonema and leafy stage. A and B may be respectively :-

- (1) *Marchantia*, *Polytrichum*
- (2) *Sphagnum*, *Funaria*
- (3) *Sphagnum*, *Polytrichum*
- (4) *Funaria*, *Polytrichum*

115. Bryophytes have _____ and _____ gametophytic generation :-

- (1) Photosynthetic, independent
- (2) Non-photosynthetic, independent
- (3) Photosynthetic, dependent
- (4) Non-photosynthetic, dependent

116. Select the correct statement :-

- (1) Liverworts asexually reproduce by fragmentation and gemmae formation.
- (2) Protonema stage is found in *Riccia*.
- (3) *Funaria*, *Sphagnum* and *Marchantia* are mosses.
- (4) Bryophytes are truly terrestrial

117. What is the fate of spore in mosses ?

- (1) It develops into thalloid body
- (2) It develops into protonema stage
- (3) It develops first into leafy stage, then protonema stage
- (4) It degenerates

118. During differentiation the spermatids remain associated with

- (1) Leydig's cells
- (2) Kupffer's cells
- (3) Spermatogonia
- (4) Sertoli cell

119. In gametogenesis, reduction division take place during :-

- (1) Multiplication phase
- (2) Growth phase
- (3) first maturation division
- (4) Second maturation

120. Longest phase of spermatogenesis is :-

- (1) Multiplication phase
- (2) Growth phase
- (3) Maturation phase
- (4) Germinal phase

121. During spermatogenesis how many spermatozoa are formed from a single primary spermatocyte

- (1) 1
- (2) 2
- (3) 4
- (4) 8

122. Which of the following part of spermatozoan has mitochondria ?

- (1) Apical cap
- (2) Head
- (3) Middle piece
- (4) Tail

123. How many ovum is formed by one primary oocyte?

- (1) 4
- (2) 3
- (3) 2
- (4) 1

- 124.** During which stage of gametogenesis meiosis occurs?
 (1) Growth phase
 (2) Multiplication phase
 (3) Maturation phase
 (4) None of the above
- 125.** The process of spermatogenesis and oogenesis in most vertebrates are under the influence of which hormone ?
 (1) Oxytocin (2) FSH
 (3) ACTH (4) ICSH
- 126.** In human at the time of birth, female gamete arrested is the form of :-
 (1) Oogonia
 (2) Primary oocyte
 (3) Secondary oocytes
 (4) Egg
- 127.** A human female has the maximum number of primary oocytes in her ovaries -
 (1) at menopause
 (2) at puberty
 (3) at birth
 (4) early in her fertile years
- 128.** Eggs liberated from ovary in human in
 (1) Secondary oocyte stage
 (2) Primary oocyte stage
 (3) Oogonial stage
 (4) Mature ovum stage
- 129.** First meiotic division during Oogenesis occurs in:
 (1) First polar body
 (2) Second polar body
 (3) Primary oocytes
 (4) Secondary polar body
- 130.** Graafian follicle contains
 (1) Many oocytes
 (2) Many sperms
 (3) A single oocyte
 (4) Site for egg fertilisation
- 131.** In mammals, corpus luteum is found in which organ
 (1) Brain (2) Ovary
 (3) Liver (4) Eyes
- 132.** Antrum is filled with fluid and is found in
 (1) Bone-marrow of bone
 (2) Cavity of brain
 (3) Graffian follicle of ovary
 (4) Pericardium of heart
- 133.** One of the following is fibrous layer of follicle
 (1) Theca externa
 (2) Zona pellucida
 (3) Membrana granulosa
 (4) Vitelline membrane
- 134.** In case of non-fertilization, corpus luteum
 (1) Starts to degenerate
 (2) Changes to corpus albicans
 (3) Starts producing progesterone
 (4) Both (1) and (2)
- 135.** Lutein cells are found in
 (1) Primary follicle (2) Corpus albicans
 (3) Corpus luteum (4) Secondary oocyte
- 136.** Corpus luteum is
 (1) Excretory (2) Endocrine
 (3) Digestive (4) Respiratory
- 137.** Luteal phase is the other name of
 (1) Follicular phase
 (2) Proliferative phase
 (3) Menstrual flow phase
 (4) Secretory phase
- 138.** Which induces the development of corpus luteum:
 (1) LH (2) Oestrogen
 (3) FSH (4) LTH
- 139.** The process of releasing the ripe female gamete from the ovary is called
 (1) Ovulation (2) Parturition
 (3) Implantation (4) Fertilisation

- 140.** Ovulation hormone is :
 (1) FSH (2) ICSH
 (3) LH (4) Testosterone
- 141.** Yellow corpus luteum occurs in a mammals in
 (1) Heart to initiate heart beat
 (2) Skin to function as pain receptor
 (3) Brain and connects cerebral hemispheres
 (4) Ovary for secretion of progesterone.
- 142.** Progesterone level falls leading to
 (1) Gestation (2) Menopause
 (3) Lactation (4) Menstruation
- 143.** Pregnancy hormone is :-
 (1) Estrogen (2) Progesterone
 (3) LH (4) FSH
- 144.** During pregnancy, the urine of female would contain
 (1) LH (2) Prolactin
 (3) FSH (4) HCG
- 145.** Site of fertilization in mammal is
 (1) ovary (2) uterus
 (3) vagina (4) fallopian tube
- 146.** Capacitation mainly take place at :-
 (1) Ampulla (2) Isthmus
 (3) Uterine cavity (4) Vagina
- 147.** The expulsion of completely developed foetus from the uterus is known as
 (1) ovulation (2) oviposition
 (3) gestation (4) parturition
- 148.** Parturition canal in female is called :
 (1) Uterus (2) Oviduct
 (3) Vagina (4) Urethra
- 149.** Oxytocin is mainly help in :-
 (1) Sperm production (2) Child birth
 (3) Urine formation (4) Gametogenesis

- 150.** Human beings are
 (1) Ovoviviparous
 (2) Oviparous (Egg laying)
 (3) Parthenogenetic
 (4) Viviparous (Directly give birth to young ones)
- 151.** Which is correct about secondary sexual characters of male
 (1) Development of mammary gland
 (2) Presence of dense hair on scalp
 (3) Low pitched voice
 (4) Development of muscle and secondary sex organs
- 152.** Which theory can explain the origin of universe?
 (1) Cosmic panspermia theory
 (2) Theory of natural selection
 (3) Big-bang theory
 (4) Theory of special creation
- 153.** What is paramount necessity for origin of life?
 (1) Light (2) Air
 (3) Water (4) Oxygen
- 154.** Protobionts were artificially synthesized by:
 (1) Oparin (2) Miller
 (3) Pasteur (4) Darwin
- 155.** Identify the correct match from column-I and column-II:

	Column – I		Column – II
P	Origin of universe	I	3 bya
Q	Origin of Earth	II	20 bya
R	Origin of first cellular form of life	III	4.5 bya
S	Origin of first noncellular form of life	IV	2 bya

Options:

- (1) P-I, Q-II, R-III, S-IV
 (2) P-II, Q-III, R-IV, S-I
 (3) P-II, Q-IV, R-III, S-I
 (4) P-I, Q-III, R-II, S-IV

156. The temperature maintained by Miller in his experiment to prove the chemical evolution:

- (1) 25 °C (2) 800 °C
(3) 5000 °C (4) 100 °C

157. Experimental evidences in favour of chemical evolution were provided by:

- (1) Louis Pasteur (2) Haldane
(3) Stanley Miller (4) Oparin

158. First inorganic and organic molecule formed on earth were respectively :

- (1) Water and ammonia
(2) Water and methane
(3) Ammonia and amino acids
(4) Ammonia and carbon dioxide

159. See the below list and identify how many steroid hormones are given :-

FSH, Aldosterone, Thyroxine, Glucagon, Adrenaline, Nor epinephrine, Testosterone, LH, Cortisole

- (1) Five (2) Three
(3) Four (4) Six

160. Which pair of hormone is known as synergistic :-

- (1) LTH & FSH (2) LTH & GH
(3) LH & GH (4) LH & FSH

161. Which drug is obtained from *Cannabis* :-

- (1) Heroin (2) Cocain
(3) Morphine (4) Bhang

162. Tissue graft in between the genetically dissimilar individual of the same species is :-

- (1) Autograft (2) Isograft
(3) Xenograft (4) Allograft

163. After maturation the lymphocytes migrate to :-

- (1) Bone marrow (2) Thymus
(3) 1 & 2 both (4) Lymph Node

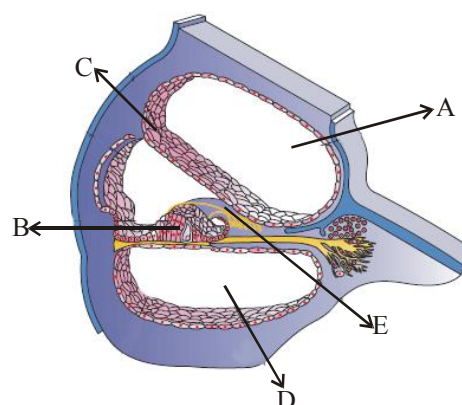
164. IgG and IgA comes through placenta and colostrum respectively are included in which immunity :-

- (1) Natural active acquired immunity
(2) Natural passive acquired immunity
(3) Artificial active acquired immunity
(4) Artificial passive acquired immunity

165. Which type of immunity is obtained by vaccines:

- (1) Active (2) Passive
(3) Non specific (4) Innate

166. Identify the option with correct description of the structures labelled as A,B,C,D and E in the adjacent figure showing sectional view of Cochlea:-



Option :-

(1)	A. Filled with perilymph	C. Separates scala vestibuli from scala tympani
(2)	B. Located on basilar membrane	D. Terminates at the oval window
(3)	B. Contains hair cells that acts as auditory receptors	E. Thin elastic membrane against which hair cells are pressed.
(4)	A. Ends at the oval window	D. Filled with endolymph

167. Match column-I with column-II and choose the correct option :

Column-I		Column-II	
A	Adrenal	I	Graves disease
B	Pancrease	II	Diabetes Insipidus
C	Pituitary	III	Acromegali
D	Thyroid	IV	Addison disease
		V	Diabetes mellitus

- (1) A-(III); B-(V); C-(IV); D-(I)
 (2) A-(IV); B-(II); C-(III); D-(I)
 (3) A-(IV); B-(V); C-(III); D-(I)
 (4) A-(I); B-(II); C-(IV); D-(III)

168. If the pineal gland of new born rat is removed then:-

- (1) It will attain sexual maturity early
 (2) Sexual maturity will delayed
 (3) It will not attain sexual maturity
 (4) Pineal gland is not concerned with sexual maturity

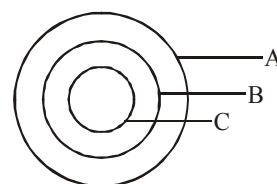
169. Consider the following four statements (A–D) and select option all the correct ones only :-

- (A) Gluconeogenesis & lipolysis stimulated by cortisol
 (B) Diabetes insipidus patients are successfully treated with insulin therapy.
 (C) When blood pressure decreased, ANF is secreted which cause dilation of blood vessels to increase B.P.
 (D) Glucagon reduces cellular glucose uptake and utilisation.

Options :-

- (1) Statement A, B & C
 (2) Statement A, D
 (3) Statement B, C
 (4) Statement A, C & D

170. A, B & C are the three layer which form wall of eyeball; choose the correct option which contribute in the formation of wall :-



- (1) A – Choroid, B – Cornea, C – Retina
 (2) A – Sclera, B – Ciliary body, C – Choroid
 (3) A – Sclera, B – Ciliary body, C – Retina
 (4) A – Cornea, B – Iris, C – Choroid

171. Match the column

Column-I		Column-II	
A.	Multipolar	(i)	Retina of eye
B.	Bipolar	(ii)	Cerebral cortex
C.	Unipolar	(iii)	Cell body
D.	Nissl's granules	(iv)	Embryonic stage

- | | | | |
|--------|-----|-----|-----|
| A | B | C | D |
| (1) ii | iii | i | iv |
| (2) i | ii | iii | iv |
| (3) ii | iv | i | iii |
| (4) ii | i | iv | iii |

172. Nerve fibres are myelinated or unmyelinated. What is true about unmyelinated nerve fibres in PNS.

- (1) Covered with oligodendrocytes
 (2) Myelin sheath is present
 (3) Layer of schwann cells is absent
 (4) Layer of schwann cells present but that do not from myelin sheath

173. Hormone X and Y both does lipolysis but x reduces the cellular uptake of glucose and promote anabolism of protein mainly while y reduces the cellular uptake of amino acid and promotes catabolism of protein mainly X and Y are :-

- (1) Insulin and cortisol
- (2) Glucagon and Somatomedin
- (3) Insulin and thyroxine
- (4) Growth hormone and cortisol

174. Given below are the name of some hormones:-

- (A) Thyroxine
- (B) Cortisol
- (C) PTH
- (D) ACTH

which of the hormones given above support erythropoiesis

- (1) C and D both
- (2) B and D both
- (3) A and B both
- (4) B and C both

175. Due to excess secretion of growth hormone in adulthood leads to :-

- (1) Gigantism
- (2) Addison's disease
- (3) Acromegaly
- (4) Exophthalmic goitre

176. The organ of corti is a structure which contains sensory cells that act as auditory receptors and is located on the -

- (1) Reissner's membrane
- (2) Tectorial membrane
- (3) Basilar membrane
- (4) Nictitating membrane

177. All of the following enzymes hydrolysed glycosidic bond except :-

- (1) Maltase
- (2) Lactase
- (3) Pepsin
- (4) Sucrase

178. Acromian process is a part of :-

- (1) Sternum
- (2) Scapula
- (3) Clavicle
- (4) Pelvic girdle

179. Find the incorrect match :-

	Column-I	Column-II	Column-III
(1)	True ribs	Vertebro-sternal ribs	14 in number
(2)	Parotid salivary gland	Largest salivary gland	1 pair
(3)	Molar teeth in adult	Ecto-endodermal in origin	12 in number in an adult
(4)	<i>Labeo</i>	Bony fish	Fresh water edible fish

180. Find the incorrect match :-

	Column-I	Column-II	Column-III	Column-IV
(1)	Sarcomere	Functional unit contraction of striated muscles	It include 1/2 I band + complete A band + 1/2 I band	Found in cardiac and skeletal muscles
(2)	Phalanges	56 in number in a normal person	2,3,3,3,3 is formula	in case of polydactyly number can be change
(3)	Amniotes	Include reptiles aves mammals	Metanephric kidney	12 pair of cranial nerves
(4)	TV + IRV + ERV = Vital capacity	ERV + IRV = expiratory capacity	ERV + RV = functional residual capacity	Total lung capacity can directly measured by spirometer

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1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

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PATTERN / TYPE : NEET / PART TEST-4
ANSWER KEY
PRE-MEDICAL 2020

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	1	1	2	3	2	2	1	1	2	2	1	3	4	2	1	3	4	1	3	4	1	1	1	2	2	2	3	4	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	4	2	3	4	1	4	3	4	4	3	2	2	4	1	4	3	3	4	4	2	1	4	4	4	2	4	4	1	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	4	2	1	4	1	3	3	2	2	2	4	2	3	3	1	1	2	1	4	2	2	2	2	3	2	2	3	3	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	3	1	1	4	3	3	1	2	3	1	1	4	1	4	1	3	2	4	4	3	1	3	1	1	1	2	4	3	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	4	3	2	2	3	1	3	3	2	3	1	4	3	2	4	1	1	3	4	4	2	4	4	4	4	3	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	3	3	1	2	2	3	2	2	4	4	4	4	2	1	3	3	1	2	3	4	4	4	3	3	3	3	2	3	4

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KOTA

SPARK COURSE - I

TARGET : PRE-MEDICAL 2020

Test Type : **PART TEST # 04**

Test Pattern : **NEET(UG)**

TEST DATE : 01 - 03 - 2020

HINT - SHEET

2. 

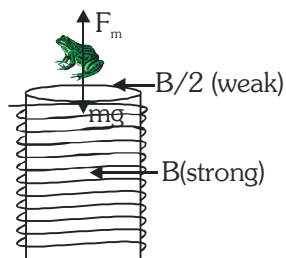
$$B = \frac{\mu_0 i}{2R}$$

$$= 8\sqrt{2} \text{ T}$$

and $B_1 = B_2 = \frac{B}{2} = 4\sqrt{2} \text{ T}$

$$\therefore B_{\text{Net}} = \sqrt{(4\sqrt{2})^2 + (4\sqrt{2})^2} = 8 \text{ T}$$

4. A stretching force or a tension force
6. Diamagnetic materials moves from stronger magnetic field region to weaker magnetic field region.
- $\Rightarrow$ Body of the frog behaves as diamagnetic.



9. For rotating conducting rod emf across its end is

$$e = \frac{1}{2} B \omega \ell^2$$

So $e_{OP} = \frac{1}{2} B \omega (4\ell)^2 = 8B\omega\ell^2$

$\therefore e_{OQ} = \frac{1}{2} B \omega (2\ell)^2 = 2B\omega\ell^2$

$$e_{PQ} = 6B\omega\ell^2$$

10. An emf will induce in that close loop in such a way that it opposes the motion of rod.

12. $e_0 = NBA\omega$

13. $i = \frac{d\phi/dt}{R} = \frac{100t}{400} = \frac{200}{400} = 0.5 \text{ A}$

14. Induced charge does not depend on time

15. Self inductance of solenoid is :-

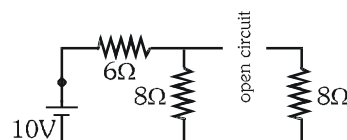
$$L = \mu_0 n^2 A \ell \quad (A = \pi r^2)$$

$$L = 4\pi \times 10^{-7} \times \frac{N^2}{\ell^2} \pi r^2 \times \ell \quad \left(n = \frac{N}{\ell}\right)$$

$$= 4\pi^2 \times 10^{-7} \times \frac{25}{2} \times 10^4 \times 4 \times 10^{-4}$$

$$L = 2 \times 10^{-4} \text{ H}$$

16. At $t = 0$



$$I = \frac{V}{R_{\text{net}}} = \frac{10}{6+8} = \frac{10}{14} = \frac{5}{7} \text{ A}$$

17. $V^2 = V_R^2 + (V_L - V_C)^2$

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{V_R^2}$$

$$V_L = V_C$$

$$V = V_R$$

$$\therefore [V_L - V_C = 0]$$

given $V = 220 \text{ V}$

$$V_R = 220 \text{ V}$$

$$i = \frac{V}{R}$$

$$i = \frac{220}{100}$$

$$i = \frac{22}{10}$$

$$i = 2.2 \text{ A}$$

OR

At resonance

$$V_{\text{applied}} = V_R$$

$$\text{so } I = \frac{V_R}{R}$$

18. $i_0 = \frac{V_0}{X_C}$ $e = 200\sqrt{2} \sin(100t)$

$i_0 = V_0 \omega C$... (i)

$V_0 = 200\sqrt{2}$; $\omega = 100$; $C = 10^{-6}F$

$i_0 = 200\sqrt{2} \times 100 \times 10^{-6}$

$i_0 = 2\sqrt{2} \times 10^{-2} A$

$i_{rms} = \frac{i_0}{\sqrt{2}} = \frac{2\sqrt{2} \times 10^{-2}}{\sqrt{2}}$

$2 \times 10^{-2} A = 20 \text{ mA}$

20. $W_{APP} = W - F_B$ (F_B = force of buoyant)

$W_{APP} = V\rho g - V\rho_1 g = V\rho g \left[1 - \frac{\rho_1}{\rho} \right]$

$W_{APP} = W \left[1 - \frac{\rho_1}{\rho} \right]$

21. When coin is removed the downward force on block decrease, which will reduce the volume submerged also the removing volume will filled by water so $h \downarrow$.

22. Equation of continuity $A_1 v_1 = A_2 v_2$

$Av = \text{constant}$

$\pi r^2 v = \text{constant}$

$v \propto \frac{1}{r^2}$

23. $v_T = \frac{2r^2 (\rho - \sigma)}{9\eta} g$

$v_T \propto r^2$ $\therefore \frac{v_{T_2}}{v_{T_1}} = \frac{r_2^2}{r_1^2}$

$\therefore v_{T_2} = v_{T_1} \left(\frac{r_2}{r_1} \right)^2 = 20 \times \left(\frac{1}{2} \right)^2 = 5 \text{ cm/s.}$

24. According to Bernoulli's theorem.

$\frac{1}{2} \rho v^2 + \rho gh + P = \text{const.}$

So if $A \downarrow$ $v \uparrow$ and $P \downarrow$
and if $A \uparrow$ $v \downarrow$ and $P \uparrow$

25. $\Delta E = 4\pi R^2 T (n^{1/3} - 1)$

$= 4\pi \frac{D^2}{4} T [(27)^{1/3} - 1] = 2\pi D^2 T$

26. $h = \frac{2T}{r\rho g} \Rightarrow h \propto \frac{1}{r}$

27. $P_1 - P_2 = \frac{4T}{r}$

$\Rightarrow \frac{4T}{r_1} - \frac{4T}{r_2} = \frac{4T}{r} \Rightarrow r = \frac{r_1 r_2}{r_2 - r_1}$

$r = \frac{r_1 r_2}{r_2 - r_1} = \frac{4 \times 5}{5 - 4} \text{ cm.} = 20 \text{ cm.}$

28. $(P_{ex})_1 = 2(P_{ex})_2$

$\frac{4T}{r_1} = 2 \left(\frac{4T}{r_2} \right) \Rightarrow \frac{r_1}{r_2} = \frac{1}{2}$

$\frac{V_1}{V_2} = \left(\frac{r_1}{r_2} \right)^3 = \left(\frac{1}{2} \right)^3 = \frac{1}{8}$

29. $h = \frac{2T \cos \theta}{\rho g r}$

As r, h, T are same, $\frac{\cos \theta}{\rho} = \text{constant}$

$\Rightarrow \frac{\cos \theta_1}{\rho_1} = \frac{\cos \theta_2}{\rho_2} = \frac{\cos \theta_3}{\rho_3}$

As $\rho_1 > \rho_2 > \rho_3$

$\Rightarrow \cos \theta_1 > \cos \theta_2 > \cos \theta_3 \Rightarrow \theta_1 < \theta_2 < \theta_3$

As water rises so θ must be acute

So, $0 \leq \theta_1 < \theta_2 < \theta_3 < \pi/2$

30. $P_{\text{inside}} = P_{\text{atm}} + \frac{2T}{R}$

$= 1.013 \times 10^5 + \frac{2 \times 70 \times 10^{-3}}{1 \times 10^{-3}} \text{ Pa}$

$= 1.013 \times 10^5 + 140 \text{ Pa}$

$= 1.013 \times 10^5 + 0.0014 \times 10^5 \text{ Pa}$

$= 1.0144 \times 10^5 \text{ Pa}$

31. Net horizontal force = Avg. pressure $\times$ Area

$= \frac{\rho gh}{2} \times \sigma h = \frac{h^2 \sigma \rho g}{2}$

32. (F) vertical surface = F_{bottom}

$$\left(\frac{h\rho g}{2}\right) \times 2\pi \frac{d}{2} h = h\rho g \times \frac{\pi d^2}{4}$$

$$h = \frac{d}{2}$$

33. GPE of body is $|U| = \frac{GMm}{r}$

Force at that point is $F = \frac{GMm}{r^2}$

$$\frac{U}{F} = r \Rightarrow F = \frac{U}{r}$$

34. Speed of satellite $V = \sqrt{\frac{GM}{r}}$

$$\Rightarrow \frac{V_B}{V_A} = \sqrt{\frac{r_A}{r_B}} = \sqrt{\frac{4R}{R}} = 2$$

$$\Rightarrow V_B = (3V)(2) = 6V$$

35. Work done = Area = $\pi \left(\frac{P_2 - P_1}{2}\right) \left(\frac{V_2 - V_1}{2}\right)$

$$= \frac{\pi}{4} (P_2 - P_1)(V_2 - V_1)$$

36. As $\lambda_1 > \lambda_2 > \lambda_3$ so $T_1 < T_2 < T_3$

37. At constant V, $P \propto nT$

so $P \propto mT$

$$\frac{P_0}{P_0/2} = \frac{32(400)}{m(300)} \Rightarrow m = \frac{64}{3} g$$

$$\text{Mass of leaked gas} = 32g - \frac{64}{3}g = 10.7g$$

38. AB : $P = \text{constant} \Rightarrow V \propto T \Rightarrow T \uparrow V \uparrow$

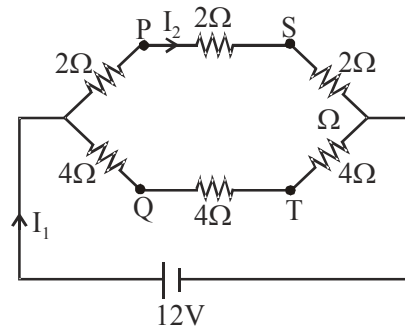
BC : $T = \text{constant} \Rightarrow V \propto \frac{1}{P} \Rightarrow P \downarrow V \uparrow$

CD : $P = \text{constant} \Rightarrow V \propto T \Rightarrow T \downarrow V \downarrow$

DA : $T = \text{constant} \Rightarrow V \propto \frac{1}{P} \Rightarrow P \uparrow V \downarrow$

39. $\therefore \frac{2}{4} = \frac{2}{4} = \frac{2}{4}$

so remove 1Ω resistor from circuit

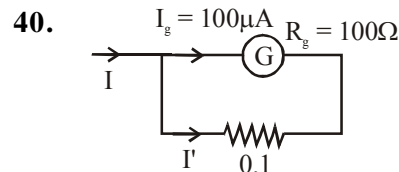


$$R_{eq} = \frac{6 \times 12}{6 + 12} = 4\Omega$$

$$\Rightarrow I_1 = \frac{12}{4} = 3A$$

$$\Rightarrow \text{Current in PQ} = 0$$

$$\Rightarrow I_2 = \frac{12}{18} \times 3 = 2A$$



$$(100\mu A) \times 100 = 0.1 \times I'$$

$$I' = 0.1 \text{ amp}$$

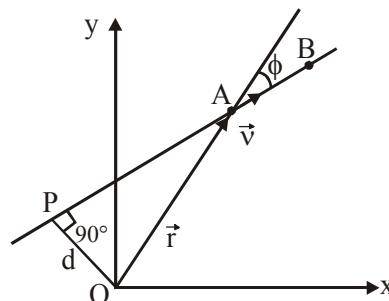
$$I = I_g + I'$$

$$= 100 \times 10^{-6} + 0.1$$

$$I = 100.1 \text{ mA}$$

43. From the definition of angular momentum,

$$\vec{L} = \vec{r} \times \vec{p} = rmv \sin \phi (-\vec{k})$$

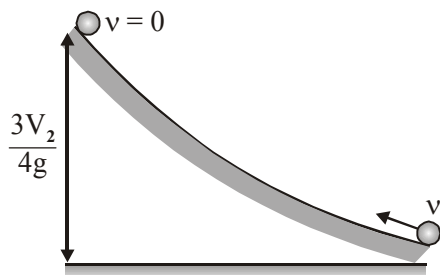


Therefore, the magnitude of L is

$$L = mvr \sin \phi = mvd$$

where $d = r \sin \phi$ is the distance of closest approach of the particle to the origin. As d is the same for both the cases, hence $L_A = L_B$.

44. From law of conservation of mechanical energy
 $\Delta K + \Delta U = 0$



$$\left[0 - \left(\frac{1}{2} I \omega^2 + \frac{1}{2} m v^2 \right) \right] + \left(m g \times \frac{3v^2}{4g} \right) = 0$$

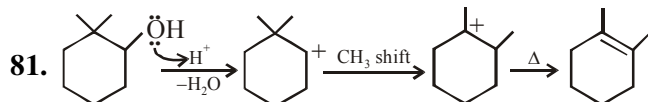
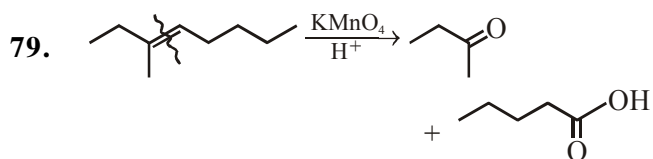
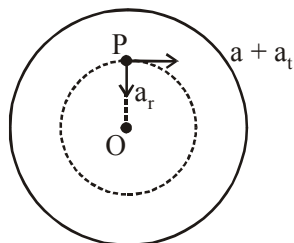
$$\Rightarrow \frac{1}{2} I \omega^2 = \frac{3}{4} m v^2 - \frac{1}{2} m v^2 = \frac{m v^2}{2} \left(\frac{3}{2} - 1 \right)$$

As body is rolling $\omega = \frac{v}{R}$

$$\text{or } \frac{1}{2} I \frac{v^2}{R^2} = \frac{m v^2}{4} \quad \text{or } I = \frac{1}{2} m R^2$$

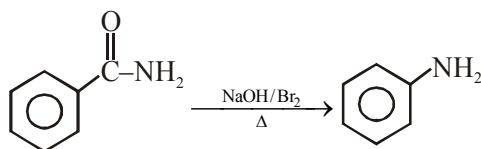
Hence, object is a disc.

45. Linear acceleration = a
 tangential acceleration = αr
 centripetal acceleration, $a_r = \omega^2 r$

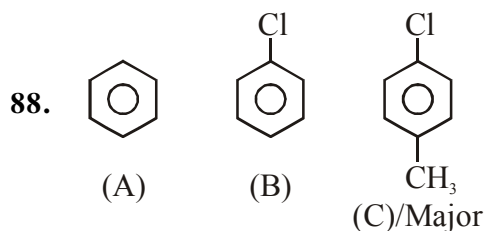
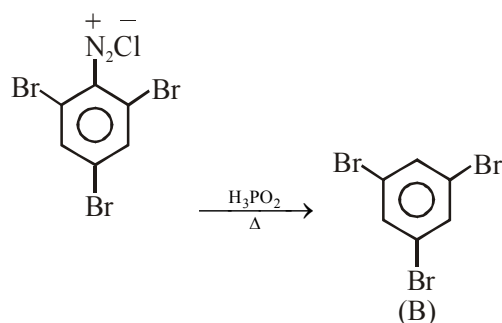
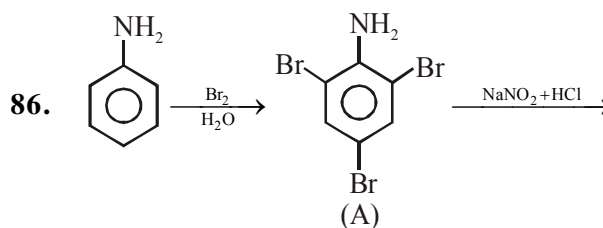


83. β -keto acid give decarboxylation at maximum rate.

84. Hoffman bromamide reaction



85. $\text{R-CN} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) SnCl}_2 + \text{HCl}} \text{R-CHO}$ stephins reduction.



91. NCERT Pg. # 8
 92. NCERT Pg. # 8
 95. Module 4 Pg. # 8
 96. A, B correct
 97. NCERT Pg. # 30
 98. Module 4 Pg. # 14
 99. Module 4 Pg. # 33
 103. A, D, E correct
 104. NCERT Pg. # 24
 105. Module 4 Pg. # 50
 107. Module 4 Pg. # 46
 110. Module Pg. # 56
 111. NCERT Pg. # 31
 113. NCERT Pg. # 31
 115. NCERT Pg. # 35
 116. NCERT Pg. # 35
 117. Module Pg. # 65